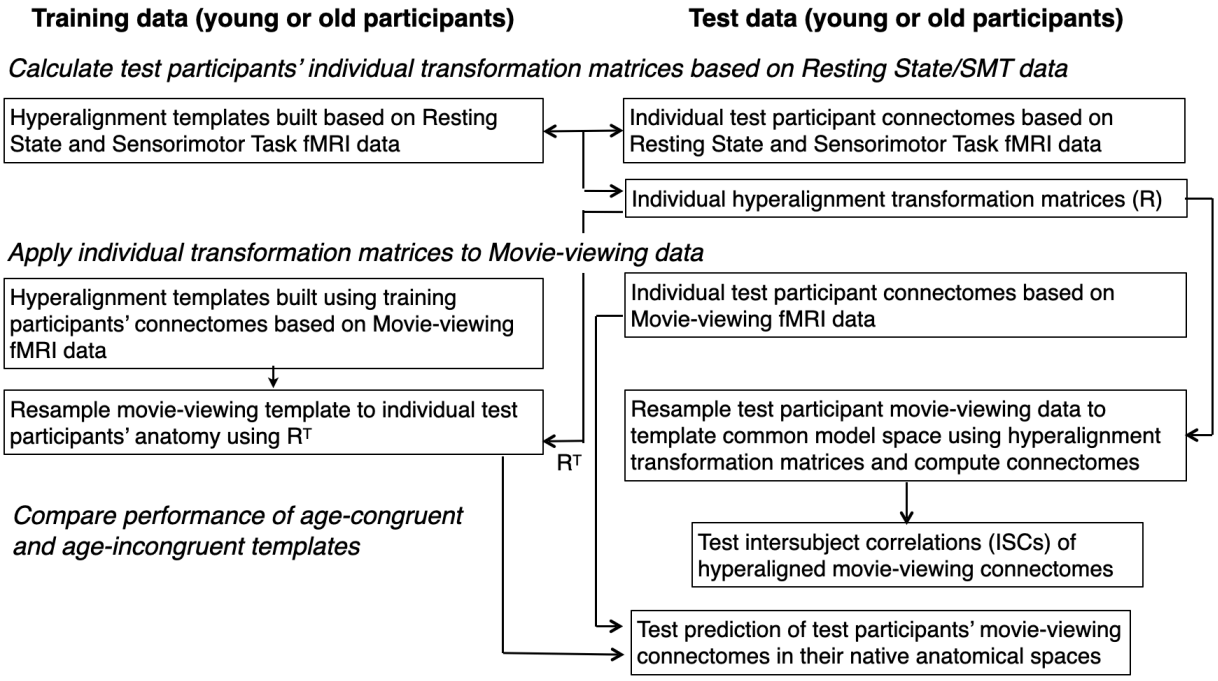
Supplementary Materials

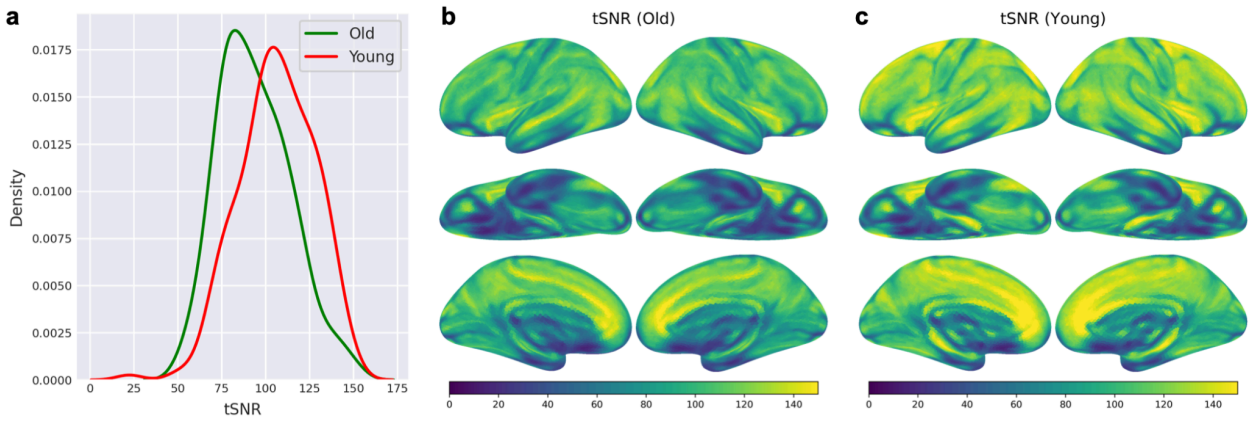
Procedure building hyperalignment templates



Supplementary Figure 1: Schematic of the procedure for building and testing hyperalignment templates

Group differences in data quality

In the main analysis, we found that both ISC and prediction accuracy were higher for the young group than the old group, regardless of template type. We suspect that this might be related to data quality differences between the groups. Therefore, we performed an additional analysis, computing the temporal signal-to-noise ratio (tSNR) as a data quality measure, and we compared it between the two groups. We computed tSNR as the mean signal divided by the standard deviation for each participant and each vertex, using the resting-state data from the Cam-CAN dataset. We found that in general the average tSNR for the young group (mean ± standard deviation = 106.4 ± 21.2) is higher than the old group (93.9 ± 20.3), $t(429) = 6.25$, $p < 10^{-9}$.



Supplementary Figure 2: tSNR differences between age groups. (**a**) The distribution of tSNR across participants, separately for each age group. (**b** and **c**) The distribution of tSNR across the cortex for the old group (**b**) and the young group (**c**).

Predicting brain responses to the movie

In the main text, we have demonstrated that individualized connectome can be better predicted using congruent templates (i.e., the same age group) than incongruent templates. In this supplementary analysis, we extend the original analysis and demonstrate that congruent age-specific templates also better predict neural responses to the movie than incongruent templates.

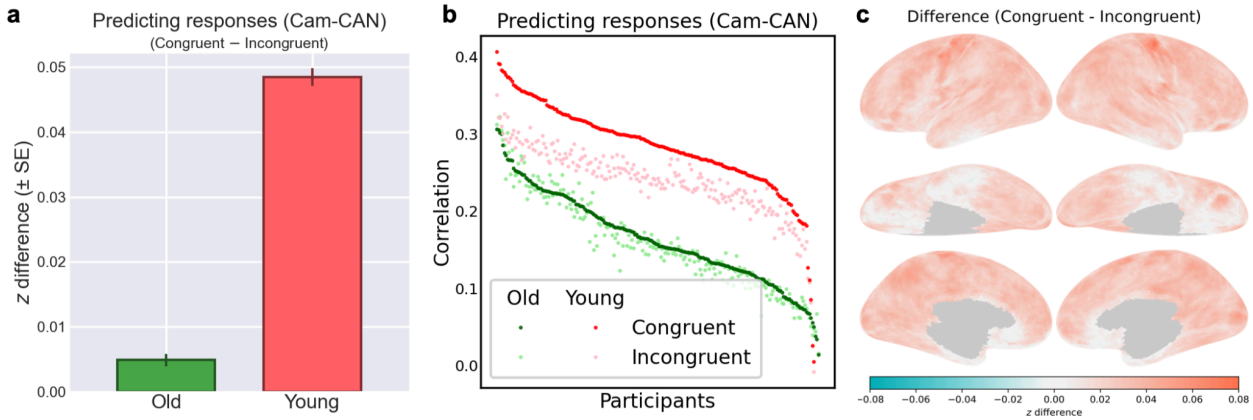
For each age group, we built functional templates using connectomes from two-thirds of the participants in each age group. The functional connectomes were based on the resting state and sensorimotor data and, thus, independent from the validation data—brain responses to the movie. With the Individualized Neural Tuning (INT) model (Feilong et al., 2023), each individual's connectome is modeled as the same template connectome with an individualized transformation, and we established correspondence between the modeled connectome and the brain responses to the movie, using the same group of training participants we used to build the template. In other words, we used the INT model to enrich the template, so that it can predict individualized brain responses in addition to individualized connectomes based on the template.

Besides using independent fMRI data, we also ensured that the participants used to validate the model were independent from the training participants (i.e., the remaining one-third of participants in the age group ("congruent"), and participants from the other age group ("incongruent")). Similar to the main analysis, we built 10 functional templates for each age group, based on different participants that were randomly chosen, and we focused on the differences in performance between using congruent and incongruent templates.

We evaluated the template performance using correlation coefficients between the measured response time series to the movie and the predicted time series. The correlation coefficients were Fisher-transformed to *z* and aggregated either across different parts of the brain (Supplementary Figure 3, panels **a** and **b**) or across participants (Supplementary Figure 3, panel **c**).

The average *z* (± SD) for the old group was 0.1661 ± 0.0600 for congruent templates, and 0.1612 ± 0.0636 for incongruent templates. The mean difference (± SE) was 0.0049 ± 0.0010, $t(212) = 5.02$, Cohen's $d = 0.3443$ and $p < 10^{-5}$. The average *z* (± SD) for the young group was 0.2925 ± 0.0627 for congruent templates, and 0.2441 ± 0.0490 for incongruent templates. The mean difference (± SE) was

0.0485 ± 0.0013, $t(209) = 35.99$, Cohen's $d = 2.4833$ and $p < 10^{-90}$. The prediction performance for both groups were better based on congruent templates than incongruent templates, with the young group showing larger congruency effect than the old group.



Supplementary Figure 3: Predicting brain responses to the movie based on congruent and incongruent templates. (a) The average z (± SD) for the old group was 0.1661 ± 0.0600 for congruent templates, and 0.1612 ± 0.0636 for incongruent templates. The mean difference (± SE) was 0.0049 ± 0.0010, $t(212) = 5.02$, Cohen's $d = 0.3443$ and $p < 10^{-5}$. The average z (± SD) for the young group was 0.2925 ± 0.0627 for congruent templates, and 0.2441 ± 0.0490 for incongruent templates. The mean difference (± SE) was 0.0485 ± 0.0013, $t(209) = 35.99$, Cohen's $d = 2.4833$ and $p < 10^{-90}$. (b) Scatter plot of each participant's average predicted response values derived from different congruent and incongruent templates. (c) Topographic distribution of predicted response difference across different brain regions.